Contents

1 Biological Background

DPP4 is an enzyme that breaks down incretine hormones, like GLP1, which are involved in glucose regulation.

- GLP-1 is a hormone which lowers blood sugar levels by stimulating insulin secretion
- Supresses the Glucagon Secretion
- Glucagon Like Peptide plays a major role in regulating blood sugar levels
- Inhibitin DPP-4 allows the GLP-1 hormone to stay activee in the body longer

1.1 Structure and Binding Side

- Type II Transmembrane Serine Protease
- Homodimer in humans

Characteristics of the Catalytic Side

Glutamic Acid = GLU

Arg = Arginine

- Serine 630 as Catalytic Centre
- Substrades are anchored by interations of GLU 205 and GLU 206
- Arg 135 has hydrogen bond interactions with carbonyl
- Phenylalanine performs π stacking

All Inhibitors have small hydrophobic moieties that can occupy the S1 pocket and hydrophlic groups that engage with the Glu205, Glu206, Arg125

2 Compound Data Aquisition

IC50 Value

The IC50 value, or half-maximal inhibitory concentration, is a measure of how much of a substance is needed to inhibit a biological process by 50%_. It's commonly used to determine the potency of an antagonist drug in pharmacological research. A lower IC50 value generally indicates a more potent drug.

- Mean of $pIC50$ lies at about 6
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3 Molecular Filtering

Lipinskis Rule of 5:

- Molecular Weight less than 500 Daltons
- Log P should be less or equal to five
- Hydrogen bond Donorns no more than 5 hydrogen bonds
- No more than 10 hydrogen bond acceptors

TPSA = Topological Polar Surface Area

Pains (Pan Assay Interference Compounds)

- Components that give false positive results in high througput screens
- Different kind of disruptive functional groups

3.1 Fingerprints

Quantitative Structure-activity relationship (QSAR) models are a valuable tool for predicting the activities and properties of query chemicals, which can help to aboid time-consuming and labour intensive experimental work

- Morgan Fingerprints represents molecular structures as binary vectors with positions or bits, where the number of each bit indicate the presence or absence of specfic atom groups in the molecule.
- Encodes a presence and the size if a atom group

Structural keys

- Encodes a molecules structure into binary vectors based on predefined structural features

Hashed Fingerprint

- Hashed fingerprints dont rely on any predefined structural features
- Enumerate all possible fragments in a molecule and convert them into numeric values with a hash function.

3.2 Difference between Dice and Tanimoto

Tanimoto Index

- Tanimoto normally gives lower score
- Tanimoto penalizes unmatched bits more, is biased against comparing small molecules with larger ones

Dice Index

- Dice tends to produce higher similarity values, especially when overlap is smalll
- Dice is better when comparing molecules of different sizes

3.3 Enrichment Plots

X-Axis (Ranked Dataset Percentage):

- Presents ratio of top ranked molecules in whole dataset
- 0 % no molecules ranked → 100 % all molecules ranked
- **Y-Axis (True Actives Identified Percentage)**:
- represents ratio of active molecules in the dataset

  Shows how the method prioritizes active compounds at different levels of ranking

We want

- Steep initial slope, high proportion of true actives should be identified
- Plateu where the ranking levels off
- Morgan Performs worse than random curve
- MACCS Keys are better in this regard

Enrichment Factor

Accesses a methods sucess with a single number rather than a plot.

- Identify active moleules in the top x% of ranked molecules

$EF(x\%) = \frac{\%\ of\ actives\ found\ at\ x\%\ of\ dataset}{x\%}$

4 Docking

- Glide Explores Conformational, orientational and positional space of the docked ligand.
- Rough position is calculated then energy optimization on OPLS-AA non-bonded potential is performed
- Uses a series of hierachical filters to search for possible locations of the ligand in the active-site region of the receptor
- Pose is a complete specification, position and orientation relative to the receptor

Functionalities of the ligands from docking

- Hydrophobic moiety in binding pocket stabilized by hydrophobic interactions
- Amine Group interacts with Glu 205, Glu 206

5 Clustering

5.1 Sillhouette Score

- Metric used to evaluate the quality of clusters in data clustering

  For a datapoint $i \in C_I$ (in cluster I) one calculates:

$a(i) = \frac{1}{|C_i|-1} \sum_{j \in C_i, i \neq j} d(i,j)$

- Basically calculates the mean distance between i and all other data points
- Because of Still high dimensionality it is difficult to achieve higher values
- Ranges from -1 to 1

5.2 Ellbow Method

The elbow method is a heuristic used in cluster analysis to determine the optimal number of clusters in a dataset. It involves plotting the explained variance as a function of the number of clusters and visually identifying the "elbow" of the curve, which suggests the point where adding more clusters doesn't significantly improve the model fit

5.3 PCA Workflow

- Data points lie as cloud in p-dimensional cartesian space
- Sum of Squares of Errors gets minimized by first principal component
- Next PCA is orthogonal to the first vector

Data matrix X is $n \times p$ each row is a repetition of an experiment, p is a feature.

l dimensional vectors of weights $w_{(k)} = (w_1, ..w_p)$ maps the row vectors to new component scores

$T = X * W$

1. Standardize $Z = \frac{X-\mu}{\sigma}$
2. Calculate covariance matrix, measures the variablitiy between two or more variables $cov(x1, x2) = \frac{\sum_{i=1}^{n}(x1_i - \bar{x1})(x2_i - \bar{x}2)}{n-1}$
3. Calculate Eigenvalues of Covariange Vectors $AX = \lambda X$

5.4 K-Means Clustering

- Unsupervised learning method for clustering data points.
- Iteratively divides data points into K clusters and minimize the variance in each cluster
- Data center is the arithmetic mean of al data points in the cluster

1. Guess some clusters
2. E-Step: Assign points to nearest clusters

3. M-step: set cluster centers to mean
4. Repeat

Cluster 2 and Cluster 3 show highest pIC50 value in the median

Important structural features:

- Pyrrolidine Ring system
- Long C-H Chain which is needed to fit in binding pocket
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6 Machine Learning

6.1 Random Forest Classifier

- Random forest are a popular supervised machine learning algorithm
- Used for solving regression and classification method
- Ensemble method, combine predictions from other models
- Smaller models in random forest ist decision tree
- Multiple decision trees are generated from subsets of data
- Predictions are made by calculating the prediction for each decision tree and then the most popular result
- n_estimators: Number of decision trees in the forest
- Splitting Metric is embedded in the criterion

Accuracy

Accuracy is **a metric that measures how often a machine learning model correctly predicts the outcome**. You can calculate accuracy by dividing the number of correct predictions by the total number of predictions.

$\frac{TP+TN}{TP+TN+FP+FN}$

Recall (True Positive Rate or Sensitivity)

n machine learning, recall (also known as true positive rate or sensitivity) *measures how well a model identifies all relevant instances (true positives) within a dataset*. It essentially quantifies the ability of a model to avoid missing any positive instances, focusing on minimizing false

$\frac{TP}{TP+FN}$

Precision

Precision is one indicator of a machine learning model's performance **the quality of a positive prediction made by the model**. Precision refers to the number of true positives divided by the total number of positive predictions

$\frac{TP}{TP+FP}$

Specificity

In machine learning, specificity refers to a model's ability to correctly identify negative instances. It's essentially the true negative rate, indicating how well the model avoids misclassifying actual negative cases as positive (false positives).

$\frac{TN}{TN+FP}$

F1-Score

The F1-score is a metric used in machine learning to evaluate the performance of a model, particularly in classification tasks . It balances precision and recall, providing a single value that represents how well the model performs when both are considered important.

- Combines Precision and Recall Score

Some Facts about RF Model

- Accuracy is stable
- Precision Increases
- Gap between train and test performance is a classic sign of overfitting
- Low standard deviation so consistent performance